

Citetax

Sri Lanka Tax Filing Copilot

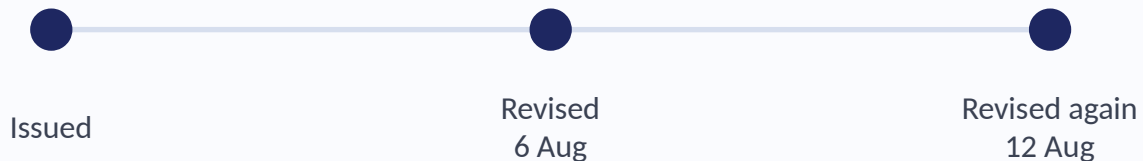
Every number, cited.

IT3041 · Mid Evaluation

Why Tax

3

versions of the same IRD circular in one week,
August 2026



A normal chatbot cites whichever copy it happened to index, with no date and full confidence.

Who needs it

Salaried employees

Does APIT already cover me, or must I still file?

Freelancers

Have I crossed the threshold? Nobody withholds for me.

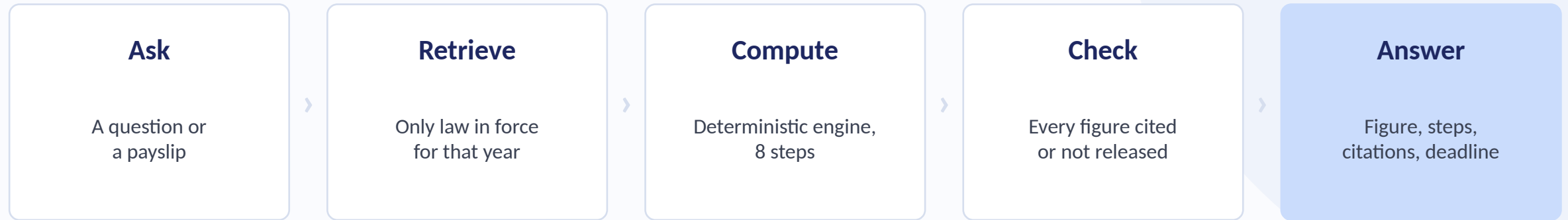
Mixed income

How do salary, interest and freelance income add up?

Four different jobs: find the law for the right date, compute, check deadlines, refuse safely. That needs agents, not one prompt.

What the System Is

Personal income tax, two years of assessment. Anything else is refused, not guessed.



What a user can ask

- 1 Do I need to file?**
Threshold and obligation, with the rule behind it
- 2 What do I owe?**
Eight steps, each with the rule it applied
- 3 What changed this year?**
Both years compared, and the amendment that did it
- 4 Out of scope questions**
Refused with a reason, such as VAT or company tax

The Stack

Interface

Next.js, TypeScript, Tailwind

Services

FastAPI, LangGraph orchestrator, agents over HTTP, MCP tools

Models

Groq gpt-oss-120b for language, BGE-M3 embeddings, BGE reranker, spaCy NER

Data

PostgreSQL 16 with pgvector, BM25 index, versioned rules table, Redis

Run

Docker Compose, Caddy TLS, GitHub Actions, single VPS

No LLM in the computation path

The model reads the question and writes the explanation. Rates come from a verified table and the arithmetic is plain Python.

Agent Flow

Orchestrator

Reads the question, fills missing facts, calls the others, builds the answer

Typed messages over HTTP, not free text

Retrieval

Hybrid BM25 and dense search, filtered to the right year

Owns: Superseded law

Computation

Deterministic engine over the verified rules table

Owns: Untraceable numbers

Compliance

Must you file, by when, at what penalty

Owns: Missed deadlines

Guardrail

Checks scope, citations and personal data before release

Owns: Uncited claims

Why five services and not one agent with tools: each one is tested on its own, so when a number is wrong we know which stage produced it. Every hop is logged.

Implementation Journey: Build Order

From trusted rules to integrated services, up to Mid Evaluation



Implementation logic

Each phase creates a reliable input for the next: verified rules → deterministic computation → comparison → compliance → integration.

Next Implementation Phases

How we move from a working core to a robust deployable system

01

NLP

Payslip NER + query expansion

02

Compliance

Penalties + instalments

03

Security

Auth + encryption + injection defence + PII masking

04

Evaluation

Larger corpus + golden-set expansion

05

Deployment

Production + smoke tests

WHY THESE PHASES?

NLP

Understand realistic user inputs better.

Compliance

Cover more filing and payment requirements.

Security

Protect taxpayer data and strengthen defenses.

Evaluation

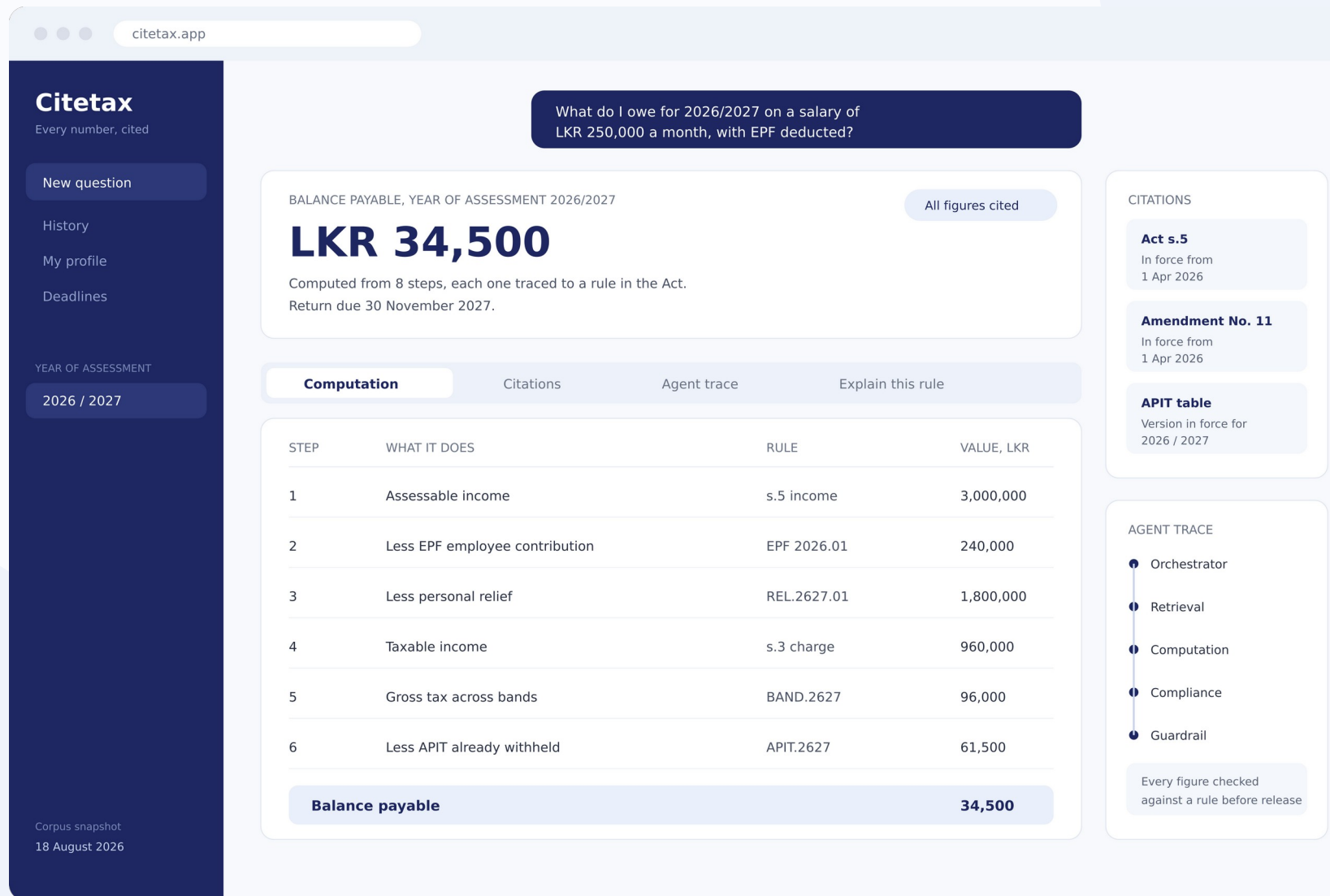
Measure reliability with broader test coverage.

Deployment

Prove the system works reliably in production.

Goal: move from a working core → evaluated → secure → deployable system

Up Next



Responsible AI

A privacy layer in front of everything

A local model checks anything a user sends before it goes further. It strips personal details such as TIN, NIC and employer name, and blocks wrong or sensitive content from reaching the hosted model. Nothing private leaves the machine.

No made up figures

The model never touches the arithmetic. Every step has a rule behind it.

No outdated law

Retrieval is filtered by year, and the citation shows the dates it applied.

No advice

Out of scope and advisory questions are refused with a reason.

Nothing hidden

The user can open every step and every source, and check the answer.

Pricing

Free

LKR 0

always

Anyone checking if they must file

- Obligation check
- One estimate per year
- Deadline reminders

Individual

LKR 1,500

per filing season

Salaried filers and freelancers

- Unlimited computations
- Payslip upload
- Year on year comparison
- Full computation export

Practice

LKR 7,500

per seat, per month

Small accounting practices

- Multi client workspace
- Bulk checking
- Branded exports
- Full audit trail

Market

Individual filers and freelancers, plus small accounting practices

Deployment

Web first, containerised, one small VPS, so a low price still works

Why they pay

A practice cannot bill for an answer it cannot check